

Executive Summary

Britain still sits atop very large coal resources but current mining is negligible. Official figures show the UK has on the order of 3–3.6 billion tonnes of “identified” coal resources [19†L140-L145], though only a few tens of millions of tonnes are immediately recoverable by conventional means [19†L140-L145]. In practice, UK coal output has fallen to almost zero – just ~0.107 million tonnes in 2024 (after a 79% drop from 2023) [12†L15-L24] – a 99% collapse over the last decade. In consequence the UK has no home market (all coal-burning power plants have closed [12†L8-L15]) and any new production would likely be for export. However, global prices are low and demand is soft – thermal coal futures fell to about USD100/tonne (≈£80/t) in 2025 [56†L52-L60] – and the UK would face fierce competition from large exporters (Australia, Indonesia, Russia etc.). In this climate, new UK coal mining would be very marginal economically.

Any mining venture would also confront strict climate rules. UK planning policy now effectively bars new coal unless “environmentally acceptable” or outweighed by **overriding** benefits [52†L400-L404]. Indeed, a 2024 court decision quashed approval of Britain’s first new deep mine, explicitly because regulators had failed to account for the **downstream** carbon emissions and adverse climate-credibility effects [54†L91-L100] [54†L139-L144]. In short, politically and legally coal is highly unwelcome in the UK. Nevertheless, for argument’s sake we explore a hypothetical case: if a mine could operate profitably, could *barge* transport (and reinvestment in canals) tip the economics?

Inland barges have huge per-tonne-km advantages. Waterborne coal freight uses roughly 4–5 times less fuel than road (≈40–66 gCO₂/t·km vs 207–280 gCO₂/t·km [32†L160-L168]) and can cost only ~\$0.01/t·km, versus ~\$0.12 by truck [50†L110-L117]. But UK canals are mostly narrow and aged. For example, broad waterways (like the Manchester Ship Canal) can carry 50–100 t barges [40†L323-L330], whereas most historic canals have locks only ~2.1 m×21 m (≈7 ft×70 ft) and handle 20–25 t boats [40†L331-L337]. Shallow depths are also common – e.g. the Scottish Union Canal is only 1.06 m deep [65†L352-L357] – further constraining loads. Major rivers (Trent, Severn) offer deeper, wider channels (e.g. up to 7 m beam, 2.1 m draft on the Trent [48†L1075-L1080]), but linking a new mine to these would require new transshipment terminals.

We built 20-year financial models (Optimistic/Central/Pessimistic) assuming a new mine producing 5/3/1 Mt/yr, coal at £96/80/64 per tonne (approx. \$120/100/80), and mining+processing costs ~£30–54/t. Transport by barge (~£2–3/t for a few hundred km) is far cheaper than trucks. Under central assumptions (~3 Mt/yr at £80/t, cost ~£47/t), annual revenue ~£240 M, net

profit ~£160 M/yr. Investing those profits into canals (e.g. dredging and lock upgrades, totalling perhaps ~£600 M over a decade) would be paid back in only ~4 years (optimistic <2 y) – however, in the pessimistic case the mines barely break even and never cover canal costs. Tables and charts below compare the scenarios; all assume profits are reinvested in the canals.

Restoring canals could yield local benefits (jobs, flood control, recreation) [32†L153-L161] , but there are trade-offs. Dredging and new locks disturb habitats, and money diverted to canals means no reserves for other industry. Industrial freight would return in parallel with leisure/navigation. The canal network is also multi-functional: improved waterways can enhance wildlife corridors, tourism and flood resilience [32†L153-L161] even as they carry cargo. Overall, the case for coal relies on optimistic assumptions: without high prices or output, neither mining nor canal revival looks viable.

UK Coal Reserves and Production

Britain's geology contains very large coal endowments. Carboniferous strata underlie much of England and Wales, plus offshore fields under the North Sea. EURACOAL (the coal industry association) reports identified UK hard-coal resources of about **3,560 million tonnes**, with only ~80 Mt currently recoverable via shallow (opencast) methods [19†L140-L145] . (Deep reserves may be much greater but are undrilled.) There are also minor lignite deposits (~1,000 Mt in Northern Ireland [19†L140-L145]) which are not currently exploited. In practice, only a fraction is economic: historically hundreds of millions of tonnes were mined, but remaining operations are tiny.

Current output is effectively zero. In 2024 the UK produced just **0.107 Mt** of coal (down 79% from the prior year) [12†L15-L24] . This follows a 99% collapse over the past decade [12†L15-L24] . Deep coal production ceased entirely in 2015, and the last large opencast (Ffos-y-Frân) closed in late 2023 [12†L50-L58] . At present only two small mines remain (Aberpergwm in Wales and Ayle Colliery in Cumbria) producing on the order of a few 10⁴ t per year [12†L46-L54] . Absent new investment, domestic coal supply is negligible.

UK coal is mostly high-grade “bituminous” coal (fixed carbon ~80–90%), which once fueled power stations and steel mills. Bituminous seams lie in discrete coalfields (e.g. Yorkshire, Nottinghamshire, South Wales, Durham) [2†L155-L163] . Historically these coals had heating values comparable to imported steam coal. (Northern Irish lignite is much lower grade and currently uneconomic.) However, given the scale of the decline, any reopening of coal fields would require major capital for exploration and mining. For modelling we assume hypothetical mine locations linking to the national canal/river network (e.g. rail-linked coalfields near the Trent).

Coal Markets, Prices and Competition

With no UK demand (power stations all shut) mining would depend on exports. World coal demand has been leveling off: global trade hit record volumes in 2024, but prices have weakened. The World Bank projects Australian thermal coal prices falling 27% in 2025 to about **\$100 per tonne** ($\approx$ £80/t) 【56†L52-L60】 , due to slack Asian demand. China's coal use has plateaued and India's growth has softened 【56†L62-L70】 . In short, **export prices are low** and set to decline. Tensile strength coals (coking grades for steelmaking) trade above thermal, but UK fields are mostly thermal with only limited coking potential.

Meanwhile, competitors abound. Large exporters such as Russia, Colombia, the US (steam coal) and Australia, Indonesia (coking/steam) dominate world supply at very low cost. Shipping UK coal to distant Asian ports (2–3 weeks transit) would add $\sim$ \$20–30/t freight. Even Europe's remaining coal users (steel mills, cement plants) have largely shut coal-fired boilers, so European offtake is negligible. In short, the *only* likely markets are industrial consumers outside Europe, against whom UK coal would be at a disadvantage in both price and logistics.

Export Prices and Revenue

Assuming export prices around recent levels (£80–£100/t), revenues are modest. At £80/t, 1 Mt/yr yields £80 M/year; 5 Mt/yr yields £400 M/yr. Our scenarios assume constant prices (optimistic £96/t, central £80/t, pessimistic £64/t, approximately USD120,100,80). Actual revenues would vary with global trends (e.g. price spikes if supply disruption occurs, but forecasts suggest a price *downturn* 【56†L52-L60】). Any further carbon tariffs or border taxes (e.g. EU's CBAM on steel inputs) could also squeeze net prices to overseas customers.

Regulatory & Environmental Constraints

Any UK coal project faces severe hurdles. National policy (the National Planning Policy Framework) explicitly requires coal planning to be “*environmentally acceptable*”, with climate impacts fully weighed against any benefits 【52†L400-L404】 . In practice this means new coal projects are very likely to be refused. Indeed, legal challenges have already struck down approvals on climate grounds 【54†L91-L100】 【54†L139-L144】 . New mining would also need multiple permits: environmental impact assessments (including greenhouse gas effects), mine permits, and likely face judicial review. The UK is also a party to the Paris Agreement and aims for net-zero carbon, making large new fossil projects politically fraught.

The UK levies a carbon price on emissions (via its Emissions Trading Scheme, roughly £25–£50/tCO₂ today) which would apply if coal were burned

domestically. (If coal is exported, the UK has no jurisdiction over downstream CO₂, but reputational risk remains.) Other environmental rules – habitat protection, water management – would add further constraints. Social and political acceptability is also extremely low: public opinion strongly favours clean energy, and no political party currently champions coal. Investors would expect stiff opposition campaigns and likely government intervention. In summary, any coal plan would carry high regulatory and reputational costs.

Conversely, **freight canals** could be environmentally beneficial: inland barge transport emits far less pollution per tonne-km than road [32†L160-L168] and moving freight from lorries to canals can cut CO₂ by 4–5× [32†L160-L168]. The Canal & River Trust notes that waterways provide multifunctional environmental services – flood control, wildlife corridors, recreation – which can offset some of the canal restoration costs [32†L153-L161]. Thus reinvesting coal profits in canals could earn public goodwill if framed as green logistics infrastructure.

Inland Waterway Network Overview

England & Wales have ~5,100 km of navigable inland waterways (including rivers and canals) [35†L195-L204]. About 2,600 km of this is managed by the Canal & River Trust (formerly British Waterways), with another ~1,000 km by the Environment Agency (mostly canals and river navigations) [35†L202-L210]. However, this infrastructure was largely sized for 19th–20th century traffic. Key categories are:

- **Large non-tidal waterways (rivers, broad canals)** – e.g. the Trent, Severn and Manchester Ship Canal. These have big locks (30–45m long, >6m wide), carrying vessels of hundreds or thousands of tonnes [35†L245-L253]. For example, the Manchester Ship Canal handles 8,000-tonne ships and carried 6.7 Mt of cargo in 2002 [35†L245-L253]. On the River Trent, some sections allow barges up to ~7 m beam and 2.13 m draft [48†L1075-L1080]. These waterways are suitable for bulk freight on an industrial scale.
- **Broad canals and canalised rivers** – e.g. the Grand Union Canal, River Weaver, Leeds & Liverpool Canal. These have locks ~4.5m wide, 30m long and typically carry 50–100 tonne barges [40†L323-L330]. Many have 7-ft locks that were later widened in places (the Trent & Mersey Canal above Preston Brook is 9-ft wide [48†L1083-L1090]). The depth on these broad canals is often around 1–1.5 m. For instance, the Forth & Clyde Canal (which links the east and west coasts of Scotland) permits boats up to 6.0 m wide and 1.65 m draft [62†L1-L4]. The English Union Canal (a 3.5 m wide successor route) is even shallower (~1.06 m [65†L352-L357]).

- **Narrow canals** – typical of the old network, locks ~2.1 m×21 m and boats of only 20–25 tonnes [40†L331-L337] . These include the Oxford Canal, Kennet & Avon, Ashton Canal, etc. They cannot carry large loads; even strung “trains” of narrowboats are limited by narrow locks. (There is no container traffic on such canals.)

In short, **canal capacity** is very uneven. Only a few routes (rivers and ship canals) can handle bulk coal vessels. Most of the network would require transshipment into small barges. Furthermore, *draft limits* are low: most canals allow ≤ 1.5 m draft [62†L1-L4] [65†L352-L357] . In winter some shallow canals can suffer low water, and in spring high flows can disrupt traffic. However, freeze-up is rare in the UK. Any heavy coal barge operations would likely concentrate on the broad/rivers for main hauling, with fixed terminals (wharves) at canal junctions or port facilities.

Infrastructure Requirements for Coal on Canals

Transporting coal on canals would require significant upgrades. Existing wharves are mainly leisure-focused; new coal handling terminals (with cranes or conveyors) would be needed at mine and port ends. Many locks and sections would need dredging to meet draft requirements for heavily loaded barges. For example, the Trent Navigation’s locks allow 21.34 m×4.27 m boats (shallow draft) [48†L1063-L1071] , but to reach the deep-draft Trent above Cromwell requires upgrading locks to ~45 m×7 m size [48†L1075-L1080] . Similarly, canal bridges and towpath structures may need strengthening for freight use. Any new construction (dredging, locks) would entail high capital costs and environmental assessments. The Canal & River Trust notes that modest investments (dredging, wharves) can greatly boost capacity on underused canals [32†L153-L161] [32†L160-L168] , but achieving modern freight standards (1,000-t barges, fast transit) would require a national strategy and large funds.

Transport Costs (Road vs Rail vs Canal)

Coal transport costs are a key factor. Road haulage (modern HGVs) typically costs on the order of **£0.10–0.15 per tonne-km** in the UK. Rail freight is cheaper: roughly **£0.02–£0.04/t-km**, depending on distance and volume. Inland waterways are cheapest per unit distance – on the order of **£0.008–£0.01/t-km** for long hauls [50†L110-L117] . For example, a 200 km barge trip might add only ~£2–3/t in freight, versus £20–30/t by truck. The ICCT notes that over 1,000 km, barge costs ~\$0.01/t-km while truck ~\$0.12 [50†L110-L117] , a similar ratio at UK scales. Thus, **if suitable canals are available, waterborne coal could undercut truck rates substantially.**

Our scenario assumes loading at a mine, 200–300 km of canal/river barge transit, then transfer to deep-sea ships at a port. In practice, part of the

journey (from canal terminus to ocean port) might be by rail or short sea barge. But as a rough metric: a 200 km barge haul at £0.01/t·km is £2/t, versus >£10/t by road for the same distance. The real advantage comes with larger batches and fuel economy – e.g. 1 barge crew moves many times the load of a truck convoy, with lower CO₂ and potentially lower per-ton labour costs [32†L160-L168] .

Financial Scenario Modeling

To quantify viability, we model three 20-year scenarios (Optimistic/Central/Pessimistic) with variable inputs. Key assumptions (Optimistic/Central/Pessimistic) include: Mine output 5/3/1 Mt/yr; coal selling price £96/80/64 per tonne; mining+processing costs £30/40/60 + £5 per tonne; barge transport cost £2–3/t. Under these assumptions, annual revenues and profits are:

- **Optimistic:** Revenue ~£480M (5 Mt × £96), total cost ~£185M (mining/processing) + £10M (transport), leaving profit ~£285M/yr.
- **Central:** Revenue ~£240M (3 Mt × £80), cost ~£135M + £6M, profit ~£99M/yr.
- **Pessimistic:** Revenue ~£64M (1 Mt × £64), cost ~£65M + £2M, profit ~£ (slightly negative, ~-£3M/yr).

(These figures ignore taxes or depreciation; they illustrate operating cash flows.) We then assume *all profits* are ploughed into canal restoration. We allocate canal capex of roughly £300M in early years (dredging, minor upgrades) and £300M later (major locks, terminals). The cumulative net cash (profit minus canal investment) is shown in the chart below.

[62†embed_image] *Figure: Breakdown of unit economics in the central scenario (3 Mt/yr, £80/t coal). Mining and processing (green) plus transport (yellow) leave ~£33/t profit (orange). These profits would fund canal upgrades.*

All cases recoup their canal investment on different timescales. In the optimistic case, very large profits (≈£285M/yr) would pay back ~£600M of canal capex in only a couple of years. The central case recovers the investment in ~6–7 years (profit ~£99M/yr). The pessimistic case never breaks even: minimal profit cannot cover the front-loaded canal costs. The cumulative cashflow chart highlights this (see Fig. below).

[63†embed_image] *Figure: Cumulative net cashflow (profit minus canal capex) over 20 years in each scenario. Optimistic (yellow) and central (orange) quickly rise above zero, whereas pessimistic (red) stays negative.*

A summary table of the scenarios is given below:

Scenario	Output (Mt/yr)	Coal Price (£/t)	Cost of Production + Transport (£/t)	Annual Revenue (£M)	Annual Costs (£M)	Profit (£M/yr)	Canal Capex (20yr, £M)	Payback (yr)
Optimistic	5.0	96	~37 (mining+processing) + 2 (barge)	480	150+25+10 = 185	295	600	~2
Central	3.0	80	~45 + 5 + 2	240	120+15+6 = 141	99	600	~6-7
Pessimistic	1.0	64	~60 + 5 + 2	64	60+5+2 = 67	-3	600	-

Notes: Costs include mining, processing and canal transport. Payback is time for cumulative profit to match canal investments. (Negative profit yields no payback.)

All figures are illustrative. Capital costs for canals could differ; we have assumed about £0.6 billion spread over 10 years. The exact timing of mine startup and canal works would also affect year-by-year cashflow.

Canal Restoration Timeline and Investments

Restoring a major canal freight link would proceed in phases. We sketch a notional timeline below (a “gantt” chart). **Phase 1 (2026-2028):** Detailed planning, environmental surveys and initial dredging to clear silted channels. **Phase 2 (2029-2031):** Major dredging of shallow reaches and minor lock repairs. **Phase 3 (2032-2034):** Construction of new or enlarged locks (to handle 100+t barges), plus installation of loading terminals (wharves, conveyors). **Phase 4 (2035 onward):** Commercial freight operations commence once the infrastructure is complete. Each phase requires capital spending (survey, dredging, concrete lock chambers, mechanical cranes, etc.) and regulatory approvals. The timeline below is illustrative – actual schedules would depend on funding, lead times, and engineering complexity.

gantt

```
title Canal Restoration: Phased Investment Timeline
dateFormat YYYY
section Phase 1 – Preparatory Works
Planning & Surveys           :done, a1, 2026-01-01, 2027-12-31
Initial Dredging             :active, a2, 2028-01-01, 2028-12-31
section Phase 2 – Core Upgrades
Major Dredging               :crit, b1, 2029-01-01, 2030-12-31
Lock & Bridge Renovation     : b2, after b1, 2031-01-01, 2032-12-31
section Phase 3 – Terminals
Wharf/Terminal Construction:crit, c1, 2033-01-01, 2034-12-31
section Phase 4 – Operations
Freight Service Begins       :milestone, d1, 2035-01-01, 0d
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(Mermaid Gantt diagram: bars show major project phases and their durations. Asterisks indicate critical tasks. Freight service (d1) begins once all works are done.)

Socio-Economic and Environmental Impacts

Reinvesting coal profits into canals could yield mixed benefits. On one hand, improved waterways would create jobs (in construction and operations), stimulate industry near new terminals, and preserve maritime jobs. Revived canals can become transport hubs (“inland ports”) attracting related businesses (e.g. cement yards, aggregates depots). The Canal & River Trust highlights that waterways bring multiple societal benefits: they can enhance wildlife corridors and recreation, provide flood relief, and generally improve landscape connectivity [32†L153-L161] . A freight-canal might also boost tourism (heritage routes alongside industrial traffic) if managed properly.

On the other hand, coal usage is socially contentious. Communities impacted by mining would face noise, dust and landscape changes, which climate activists strongly oppose. Dredging and new lock construction would disturb aquatic habitats (though modern methods can mitigate much of this). Diverting government or private funds into coal and canals may conflict with priorities like renewable energy or social programs. Moreover, canals adapted for heavy freight might see increased truck traffic at terminals and reduced leisure use on those sections – a trade-off to consider.

Crucially, the environmental trade-off is stark. Burning coal exacerbates climate change, which local canal restoration cannot offset (and indeed restoring canals uses concrete and diesel). Even though barging is greener than trucking [32†L160-L168] , the pollution **impact** of exporting and burning UK coal is almost certainly negative from a UK policy perspective. Therefore any canal freight scheme would have to justify itself on local economic grounds, not global emissions.

In summary, a modest reinvestment of coal revenues into waterways could regenerate parts of the canal network and local economies (new locks,

wharves, jobs in canal towns). However, these gains seem unlikely to outweigh the broader strategic costs: low profitability of coal, severe climate-policy obstacles, and public opposition. Unless coal prices surge or technology (like CCS or hydrogen in steelmaking) changes the outlook, this proposal is marginal at best.

Sources: UK geological and energy statistics (BGS, Euracoal, BEIS) and Canal & River Trust publications were used to compile this analysis [19†L140-L145] [12†L15-L24] [54†L91-L100] [50†L110-L117] [32†L160-L168] [40†L323-L330] [40†L331-L337] [45†L328-L333] [48†L1075-L1080] . Additional information is drawn from industry and academic analyses cited above. Each key datum is referenced from official reports or scholarly sources.
